MC 451 Definitions Practice: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), as the student study the Project Prospectus example set up: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit. Edge cases are paraphrased from tweets in the archive; none is quoted.
- The rubric map at the end shows which grading row each part earns and why.

The example

Study	**Learning Talk and VR Tone: Affordances and Constraints in Tweets About Playing in Virtual Reality**
Unit of analysis	One English-language tweet, not a retweet, posted in 2021, that mentions VR together with play. Every variable below is coded on one tweet.

Variable 1: Learning reference

Part	**Learning reference**
Conceptual definition	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR: learning, teaching, studying, practicing, or being trained.
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person (the writer, a named person, learners in general, or a customer in an advertisement) learning, teaching, practicing, studying, or being trained in something, and the activity happens in or through VR. Otherwise mark 0. If the tweet contains a learning word, check what the word means in its sentence before marking.
Level of measurement	Nominal: two named categories with no order between them.
Coded values	0 = no learning reference. 1 = learning reference.

Part	Learning reference
Edge-case decision rules	1. A learning word that is part of a game or product title (a shooting game called a trainer, an app called a trainer) or part of an idiom (hype train, learned the hard way, learned my lesson) is coded 0 unless another clause describes learning. 2. "Learn" meaning "find out about" (learn which headset games were played most this year) is coded 0. 3. A tweet that only shares a video title is coded 1 when the title describes someone teaching or learning in VR (a video in which one avatar teaches another how to do something in a social VR app), because the unit is the tweet text and the text describes learning. 4. An advertisement for a product that teaches (a VR app that teaches you to play the piano) is coded 1. The variable records that learning is described, not who describes it or why. 5. Machine-learning and artificial-intelligence hashtags are coded 0: the learning is a computer's, not a person's.

Variable 2: Tone toward VR

Part	Tone toward VR
Conceptual definition	The writer's evaluation of VR or of the VR experience the tweet describes: whether the tweet speaks well of it, badly of it, or neither.
Operational definition	Find every word, phrase, or emoji in the tweet that evaluates VR, a VR headset, or a VR experience (fun, love, amazing; nauseous, broken, waste of money). Mark 3 when every evaluation is favorable, 1 when every evaluation is unfavorable, and 2 when the tweet has both or has none. Evaluations of something other than VR (a friend, a game that is not in VR, a public figure) do not count.
Level of measurement	Ordinal: three categories in order from negative to positive, with no claim that the steps between them are equal.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.
Edge-case decision rules	1. A tweet that praises a game played outside VR, a streamer, or a friend but says nothing evaluative about VR itself is coded 2. 2. An advertisement that lists features or a release date with no evaluative word is coded 2. An advertisement that calls the experience amazing is coded 3, because the variable records what the tweet says, not whether the writer is sincere. 3. A complaint the writer laughs off (a laughing emoji after describing a glitch that threw them out of a match) is coded by its literal words, so it is 1 unless a favorable clause also appears, which makes it 2. 4. Emoji count as evaluations: a heart or a fire emoji attached to VR is favorable, and a vomiting face is unfavorable.

Variable 3: Affordance named

Part	Affordance named
Conceptual definition	The benefit the tweet attributes to VR: something VR lets a person do, or do better, in the writer's account.
Operational definition	Look for a clause that says what VR, a headset, or a VR experience lets someone do or makes better. Match that clause to one category below. If the tweet names benefits in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a benefit, mark 0.
Level of measurement	Nominal: named categories with no order.
Coded values	0 = no benefit named. 1 = fun or engagement (enjoyment, excitement, being absorbed). 2 = hands-on interaction (moving, building, handling objects, being inside the scene). 3 = doing it with others (playing, meeting, or learning alongside other people). 4 = practicing a skill (getting better at something, in a game or outside it, in a safe setting). 5 = other benefit.
Edge-case decision rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking, a high refresh rate) is coded 0. A feature is not a benefit until the tweet says what it is for. 2. "Fun" or "love it" with no more detail is coded 1. 3. Learning to play a VR game better is coded 4, the same as practicing piano in VR, because the variable records the kind of benefit, not whether it carries outside the game. 4. A tweet that praises VR for company is coded 3 only when other people are involved; a solo experience described as absorbing is coded 1.

Variable 4: Constraint named

Part	Constraint named
Conceptual definition	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged the writer or someone else from doing something in VR.
Operational definition	Look for a clause that names a problem with VR, the hardware, or the experience that limited or discouraged use. Match it to one category below. If the tweet names problems in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a problem, mark 0.
Level of measurement	Nominal: named categories with no order.

Part	Constraint named
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, nausea, eye strain, headache, fatigue). 2 = hardware cost (the price of a headset or computer, cannot afford one). 3 = technical performance (lag, crashes, glitches, setup trouble, incompatibility). 4 = other obstacle (space, time, other players' behavior, anything else).
Edge-case decision rules	1. A problem with something other than VR (a slow phone, a friend's lost username) is coded 0 unless the tweet ties the problem to VR. 2. A warning or advice about a problem (how to build up tolerance to motion sickness) is coded in that category even when the writer did not suffer it, because the tweet names it as an obstacle. 3. Being thrown out of a match by a glitch is coded 3; what the writer lost as a result does not change the category. 4. "I can't play, no headset yet" is coded 2 only when price or affordability is stated; otherwise it is coded 4.

Rubric map

Points are the rubric's split of the 25: conceptual definition 4, operational definition 7, level of measurement 3, coded values 4, edge-case decision rules 7. Each row looks across all four variables.

Rubric row	Points	Where the example earns it	Why it earns the full row
Conceptual definition	4	The first row of each variable's table	Each one says in one or two sentences what the variable is meant to capture in the abstract, and none is a list of examples. None of the four repeats its operational definition.
Operational definition	7	The second row of each variable's table	Each one tells the coder what to do (read, find, match, mark) and how to resolve a tie, so a stranger could follow it and reach the same answer. None restates the conceptual definition in other words.
Level of measurement	3	The third row of each variable's table	Named for all four, with one phrase saying why. Each matches its values: the two-category and the category variables are nominal, and the only ordered set of values, tone, is the only ordinal variable.
Coded values	4	The fourth row of each variable's table	Every value is listed with what it means. Every tweet fits exactly one value, because each category variable carries a none value and an other value, and the operational definition says what to do when two categories apply.

Rubric row	Points	Where the example earns it	Why it earns the full row
Edge-case decision rules	7	The numbered rules in the last row of each variable's table	Four or five rules per variable, above the minimum of two. Every rule is written as an instruction a coder can apply to the next tweet, not as an intention, and each comes from a case that occurs in the archive: game titles that contain a learning word, advertisements, idioms, laughed-off complaints, features named without a purpose, and warnings about motion sickness.

Four variables done fully is the assignment. Every rule above is phrased so it could be pasted into the codebook unchanged.